

Vehicle, Car, Truck, Minivan Reflection log

Understanding the Problem

I understood the problem as creating a vehicle program where the user could choose a vehicle type and enter information like make, model, and year. The program would then display the vehicle information using inheritance.

At first, I was confused about how one method (displayVehicle) could work with different vehicle types. I learned that this works because all subclasses inherit from the Vehicle class.

Code reference:

```
60 public static void displayVehicle(Vehicle v)
61 {
62     System.out.println("Make: " + v.getMake());
63     System.out.println("Model: " + v.getModel());
64     System.out.println("Year: " + v.getYear());
65     System.out.println("City Fuel: " + v.getFuelEconomyCity());
66     System.out.println("Highway Fuel: " + v.getFuelEconomyHwy());
67     System.out.println("Seating: " + v.getSeatingCapacity());
68     System.out.println("Cargo Volume: " + v.getCargoVolume());
69 }
```

Planning the Solution

I planned the solution by first creating the Vehicle parent class and subclasses like Car, Truck, and Minivan. After that, I added user input and created a method to display vehicle details.

I used inheritance to avoid repeating code in each subclass.

Code reference (inheritance):

```
3 public class Car extends Vehicle
```

Implementation

I implemented the program step by step. First, I added user input using Scanner. Then I created objects depending on the user's choice and displayed the vehicle information using a separate method.

I tested the program by entering different vehicle choices and checking if the correct information was shown.

Code reference (user input):

```
20         System.out.print("Enter make: ");
21         String make = input.nextLine();
```

Overcoming Challenges

The most difficult part was understanding how the displayVehicle() method could accept all vehicle types. I also had trouble handling nextInt() and nextLine() together at first.

I solved this by adding input.nextLine(); after nextInt() to clear the scanner buffer.

Code reference (scanner fix):

```
16         int choice = input.nextInt();
17         input.nextLine();
```

Learning

I learned more about inheritance, polymorphism, and user input in Java. I also learned how helper methods can make programs cleaner and reduce repeated code.

Code reference (display method):

```
62         System.out.println("Make: " + v.getMake());
```